

ML and DS Internship Practical

Month 1 – Python Fundamentals

Quiz Application

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Submitted To: -

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Technology Used: Python

Development Environment: Visual Studio Code

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1. Project Title: -

Quiz Application

2 Introduction: -

A Quiz Application is a software program designed to test a user's knowledge through a series of questions and answers. This project is developed using Python and runs in the command-line interface. The application presents multiple-choice questions related to Python programming, accepts answers from the user, checks whether the answers are correct, and calculates the final score and percentage.

This project helps beginners understand how basic programming concepts such as dictionaries, loops, conditional statements, and user input can be combined to create an interactive and practical application.

3. Aim: -

To develop a console-based Quiz Application using Python that allows users to answer multiple-choice questions related to Python programming and calculates the final score and percentage.

4 Objectives: -

To understand the use of dictionaries in Python.

- To practice loops and conditional statements.
- To implement score calculation logic.
- To improve user interaction through a quiz system.
- To apply basic Python concepts in a real-world project.

5. Problem Statement: -

Develop a Python-based Quiz Application that presents multiple-choice questions about Python programming. The application should accept answers from the user, verify the responses, display whether each answer is correct or incorrect, and finally show the total score and percentage.

6. Scope of the Project: -

The Quiz Application is designed for students and beginners who want to test their knowledge of Python programming. The application can be used for self-learning, educational activities, and programming practice. The project can also serve as a foundation for developing more advanced quiz systems.

7. System Requirements: -

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: 4 GB or above
- Storage: 100 MB free space

Software Requirements:

- Operating System: Windows, Linux, or macOS
- Python: Version 3.x
- Code Editor: Visual Studio Code or IDLE
- Git and GitHub for version control

8. Features of the Application: -

- Contains 20 multiple-choice questions based on Python programming.
- Displays four options (A, B, C, and D) for each question.
- Accepts answers from the user.
- Checks whether the answer is correct or incorrect.
- Displays the correct answer if the user's answer is wrong.
- Calculates total score and percentage.
- Displays the final result after completing all questions.

9. Technologies Used: -

- Python 3.x
- Visual Studio Code
- Git
- GitHub

10. Concepts Used: -

- Variables
- Dictionary
- Lists
- For Loop
- If-Else Statements
- User Input using input()
- String Methods using upper()
- Built-in Function len()
- Arithmetic Operators

11. Methodology: -

The application stores questions, answer choices, and correct answers in a dictionary. A loop is used to display each question one by one. The user enters an answer, which is converted into uppercase and compared with the correct answer stored in the dictionary. The score is updated whenever the answer is correct. After all questions are completed, the program displays the total score and percentage.

12. Algorithm: -

Step 1: Start the program.

Step 2: Create a dictionary containing questions, options, and correct answers.

Step 3: Initialize score = 0 and question_number = 1.

Step 4: Display the quiz title.

Step 5: Display one question and its options.

Step 6: Accept the user's answer.

Step 7: Convert the answer to uppercase.

Step 8: Compare the answer with the correct answer.

Step 9:

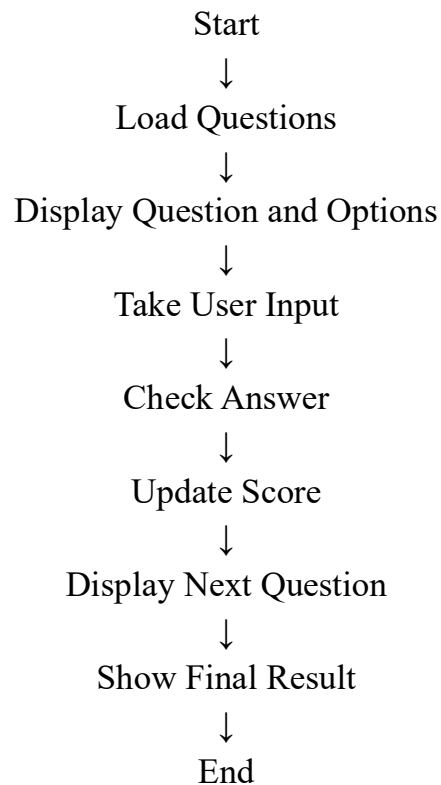
- If the answer is correct, increase the score by 1.
- Otherwise, display the correct answer.

Step 10: Repeat until all questions are completed.

Step 11: Display the final score and percentage.

Step 12: End the program.

13. Program Flow: -



14. Screenshot: -

```
PS C:\Users\Saniya Shaikh> & "C:/Users/Saniya Shaikh/AppData/Local/Programs/Python/Python38-32/Python.exe" C:/Users/Saniya Shaikh/AppData/Local/Programs/Python/Python38-32/Scripts/python.exe
===== Quiz Application =====

Question 1
1. Who created Python?
A. Guido van Rossum
B. James Gosling
C. Dennis Ritchie
D. Bjarne Stroustrup
Enter your answer (A/B/C/D): a
Correct Answer!

Question 2
2. Which symbol is used for comments in Python?
A. //
B. <!-- -->
C. #
D. /* */
Enter your answer (A/B/C/D): c
Correct Answer!

Question 3
3. Which function is used to display output in Python?
A. display()
B. print()
C. output()
D. show()
Enter your answer (A/B/C/D): b
Correct Answer!

Question 4
4. Which data type is used for decimal numbers?
A. int
B. str
C. bool
D. float
Enter your answer (A/B/C/D): d
Correct Answer!
```

```
===== QUIZ RESULT =====
Total Questions : 20
Correct Answers : 19
Wrong Answers : 1
Final Score : 19 / 20
Percentage : 95.0 %
```

15. Testing Report: -

Test_Case ID	Test Scenario	Input	Expected Output	Status
TC-01	Program starts successfully	Run the program	Quiz title is displayed	Passed
TC-02	Display first question	No input required	First question and four options are displayed	Passed
TC-03	Enter correct answer	A	Displays "Correct Answer!" and score increases by 1	Passed
TC-04	Enter wrong answer	B	Displays "Wrong Answer!" and shows the correct answer	Passed
TC-05	Enter lowercase answer	a	Converts to uppercase and checks answer correctly	Passed
TC-06	Display next question	Any valid answer	Next question is displayed	Passed
TC-07	Score calculation	15 correct answers	Final score displayed as 15/20	Passed
TC-08	Wrong answer count	15 correct answers	Wrong answers displayed as 5	Passed
TC-09	Percentage calculation	Score = 15	Percentage displayed as 75.0%	Passed
TC-10	Complete all questions	Answer all 20 questions	Final result is displayed successfully	Passed

16. Expected Output: -

The application displays Python-related questions one by one, accepts answers from the user, verifies the responses, and displays the final result including total questions, correct answers, wrong answers, score, and percentage

17. Advantages: -

- Easy to understand and use.
- Improves Python programming knowledge.
- Demonstrates basic Python concepts.
- Provides immediate feedback to the user.
- Helps beginners practice problem-solving skills.
- Can be easily modified to add more questions.

18. Limitations: -

- Questions are fixed and cannot be added during runtime.
- Supports only multiple-choice questions.
- Stores data temporarily while the program runs.
- Does not provide a graphical user interface.
- Does not maintain previous quiz records.

11 Conclusion: -

The Python Programming Quiz Application successfully demonstrates the use of dictionaries, lists, loops, conditional statements, and user

input handling in Python. The project provides an interactive way to test Python knowledge while strengthening programming fundamentals. It is an excellent beginner-level project that combines multiple Python concepts into a practical application.

12 GitHub Repository Link: -

<https://github.com/saniya2698/Quiz-Application-python/tree/main>